

Algebra II with Trigonometry

Chapter 6.2

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4506907_alg-2trig-h-chp-6-2-note-docx.pdf](#)

- 1. Unit Circle Coordinates** If the terminal point on the unit circle is $(-\frac{3}{5}, \frac{4}{5})$, what are $\cos t$, $\sin t$, and $\tan t$?
- 2. Quadrant Sign Check** If t is in Quadrant III, which of these are positive: $\sin t$, $\cos t$, $\tan t$, $\sec t$?
- 3. Reference Angle Thinking** Without finding a decimal, what is $\sin(\frac{5\pi}{6})$?
- 4. Using Periodicity** What is $\tan(15\pi)$?
- 5. Reciprocal Function** If $\cos t = -\frac{1}{2}$, what is $\sec t$?
- 6. Even or Odd** Is $\cos(-t)$ equal to $\cos t$ or $-\cos t$?
- 7. Odd Function Value** If $\sin t = \frac{3}{4}$, what is $\sin(-t)$?
- 8. Pythagorean Identity** If $\sin t = \frac{3}{5}$ and t is in Quadrant I, what is $\cos t$?
- 9. Quadrant from Signs** If $\csc t > 0$ and $\sec t < 0$, in which quadrant does the terminal side of t lie?
- 10. Trig Model of Motion** For $y(t) = 4\cos(3\pi t)$, what is the displacement when $t = 0$?